SDU - University of Southern Denmark

Peer Review:
Robust Visual Question Answering with a Small Vision-Language Model

Chapter 1

Summary of the plan in your own words

The project formulation is a joint project combining together Computer Vision and Natural Language Processing topics, evaluating a small vision-language model (*SmolVLM-500M-Instruct*) to answer image related questions. The authors plan to establish a zero-shot baseline, fine-tune the model using LoRA, and compare accuracy on a reformulated subset of the test set, determining whether the model perceives the meaning of the question rather than depending from the question structure or word choice.

Chapter 2

Strengths

The key strengths of this project formulation are:

- **Multimodal Task**.
  The project task is relevant to both Computer Vision and Natural Language Processing learning outcomes, as the proposed objective requires text and image manipulation techniques.
- **Well-Structured Progression**.
  Starting with the base model before applying fine-tuning provides a clean overview of the model capabilities. In particular, following this approach, we should be able to define whether the model is based on syntactic structures or has semantic understanding.
- **Error Analysis**.
  Incorrect predictions are divided into Visual Understanding, Language Understanding and Interactions between the two Modalities, adding strong analytical analysis that goes beyond raw accuracy numbers.

Chapter 3

Main concerns, ordered by importance

While the project formulation is well-structured and feasible, there are some concepts that may need additional explanations.

- **Rephrasing Methodology**.
  The proposal does not state how many test questions will be rephrased or how they will be generated (manually or using other LLMs), and also it does not define the degree of linguistic variation: simple synonym swaps or something more sophisticated like syntactic restructuring?
- **Metric Ambiguity**.
  The accuracy metric will be the primary evaluation of the quality of the model's text generations. However, from the proposed project formulation, it is not clear which kind of accuracy will be used. Will verbose text generations be penalized? Will you use exact-match accuracy or something like string normalization? Please consider providing a more detailed explanation of it in your final report.
- **Overfitting**.
  VQAv2 dataset consists of questions about images that often follow repetitive patterns. Fine-tuning the model on a subset of 10.000 samples may cause it to overfit: recognizing the specific question-answer patterns rather than increasing its robustness. As a result, the fine-tuned model might perform well on the test set, while its general capabilities could actually decrease.

Chapter 4

Concrete suggestions

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Chapter 5

The single most important change you'd make